

SANNICOLAU MARE, RomaniaWMO: **151990**Lat: **46.0714N**Lon: **20.6017E**Elev: **86**StdP: **100.29**Time Zone: **2.00 (EUE)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.2	-7.9	-12.5	1.3	-8.8	-10.0	1.6	-5.9	11.3	-2.6	9.9	2.1	2.0	0	0.487

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.5	34.1	21.4	32.3	21.4	30.6	21.0	24.1	28.7	23.0	28.6	22.1	28.4	3.2	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.0	17.9	26.0	21.5	16.3	24.8	20.4	15.2	23.9	72.8	28.4	68.6	28.7	65.2	28.3	28.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.1	8.5	7.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.0	-0.4	3.0	6.5	11.9	17.2	20.9	22.1	22.9	18.0	12.3	7.2	1.7
	DBStd	8.95	4.47	4.66	4.38	4.68	3.58	3.34	3.16	3.39	4.19	4.36	4.35	4.21
	HDD10.0	1067	322	197	123	34	1	0	0	0	0	27	106	257
	HDD18.3	2752	581	429	367	195	63	13	6	4	58	189	333	515
	CDD10.0	1798	0	1	14	92	224	327	377	401	239	99	22	1
	CDD18.3	441	0	0	0	3	28	91	123	146	47	3	0	0
	CDH23.3	4697	0	0	1	58	302	912	1354	1582	442	46	0	0
	CDH26.7	1817	0	0	0	5	61	318	556	729	146	2	0	0
Wind	WSAvg	3.0	2.9	3.2	3.4	3.6	3.1	2.6	2.7	2.6	2.7	2.7	3.1	3.0
Precipitation	PrecAvg	554	32	28	29	41	59	71	64	43	50	48	42	46
	PrecMax	814	58	76	111	104	156	205	149	88	129	115	98	92
	PrecMin	264	5	1	1	4	11	12	17	5	4	2	3	3
	PrecStd	137	17	19	23	24	36	45	31	25	27	27	22	27
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.8	15.9	22.0	27.2	30.2	33.8	35.6	36.8	33.2	26.4	20.1	13.2
		MCWB	9.8	11.0	13.6	17.2	20.8	23.2	22.2	21.3	20.3	16.8	14.7	10.5
	2%	DB	8.5	13.7	18.6	25.1	28.5	32.0	33.6	34.6	30.3	24.6	17.3	10.5
		MCWB	6.8	10.0	12.0	15.9	19.5	22.4	21.9	20.7	19.7	16.0	13.4	8.9
	5%	DB	7.0	11.4	15.7	22.6	26.7	30.2	31.6	32.7	28.0	22.5	15.5	8.4
		MCWB	6.1	8.5	10.2	14.5	18.6	22.0	21.3	20.7	18.4	15.1	12.9	7.3
	10%	DB	5.5	9.5	13.4	20.4	24.9	28.3	29.7	30.8	26.0	19.8	13.6	6.8
		MCWB	4.8	7.6	8.6	13.3	17.5	21.0	20.4	20.6	18.2	14.2	11.4	6.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.9	12.4	14.9	18.3	23.7	26.3	24.6	23.9	23.0	18.1	15.5	11.6
		MCDB	11.8	15.0	21.1	25.2	27.4	30.6	28.0	29.1	30.2	23.2	17.7	12.9
	2%	WB	7.4	10.7	12.8	16.9	20.8	24.6	23.6	22.9	20.7	17.0	14.3	9.3
		MCDB	8.3	12.7	17.9	23.9	25.8	28.9	28.8	29.8	27.6	22.2	16.8	10.6
	5%	WB	6.2	9.1	11.1	15.7	19.5	23.2	22.6	22.1	19.7	16.0	13.1	7.7
		MCDB	7.0	11.3	15.0	21.4	24.5	27.7	29.1	28.9	25.4	21.2	15.7	8.6
	10%	WB	4.9	7.7	9.6	14.4	18.5	22.1	21.7	21.4	18.7	14.9	11.7	6.4
		MCDB	5.7	9.2	12.9	19.5	23.1	26.8	28.6	28.2	23.6	19.2	13.5	7.0
Mean Daily Temperature Range	5% DB	MDBR	5.8	7.7	10.6	11.9	11.4	11.9	12.4	12.5	11.2	10.9	7.7	5.4
		MCDBR	7.5	11.2	14.2	15.5	14.0	13.6	14.5	15.1	14.2	14.7	10.1	7.9
		MCWBR	5.6	7.6	8.4	7.6	6.3	5.7	5.5	4.5	5.3	6.8	6.6	6.3
	5% WB	MCDBR	7.0	10.0	12.9	13.5	12.1	12.1	12.8	12.7	11.9	13.1	9.2	7.8
MCWBR		5.7	7.2	8.2	6.9	6.2	6.2	6.0	5.0	5.0	6.5	6.3	6.3	
Clear-Sky Solar Irradiance	taub	0.341	0.366	0.401	0.456	0.440	0.436	0.448	0.452	0.430	0.393	0.362	0.340	
	taud	2.342	2.281	2.215	2.065	2.182	2.225	2.189	2.183	2.213	2.310	2.349	2.357	
	Ebn at Noon	740	793	817	800	831	834	817	796	777	752	706	696	
	Edh at Noon	77	99	122	155	142	137	140	136	121	96	77	69	
All-Sky Solar Radiation	RadAvg	1.14	1.99	3.25	4.63	5.62	6.17	6.12	5.49	3.83	2.58	1.50	0.90	
	RadStd	0.12	0.29	0.43	0.44	0.52	0.48	0.47	0.51	0.42	0.27	0.13	0.13	